

Logic in Computer Science
Problem Sheet 6: Semantic Tableau
September 28, 2025

1. Prove the following formulas are valid using semantic tableau.
 - (a) $(p \rightarrow (q \rightarrow p))$
 - (b) $((\neg(p \vee q)) \leftrightarrow ((\neg p) \wedge (\neg q)))$
2. Construct a semantic tableau for the formula $(p \vee q) \wedge (\neg q \wedge \neg p)$ and conclude whether the formula is satisfiable or not.
3. Prove or disprove: If every branch in a semantic tableau for A is open, then A is valid.
4. Convert the following formulas into equivalent DNFs using semantic tableau.
 - (a) $(\neg(((\neg p) \rightarrow q) \rightarrow r))$
 - (b) $(p \vee q) \wedge (\neg q \vee r \vee \neg s) \wedge (r \vee \neg p)$
5. Convert the following formulas into equivalent CNFs using semantic tableau.
 - (a) $(p \leftrightarrow (q \vee r))$
 - (b) $((\neg p) \rightarrow q) \vee ((p \wedge (\neg r)) \leftrightarrow q)$
6. Check whether the following set of formulas are satisfiable or not using semantic tableaux.
 - (a) $U = \{(p_3 \rightarrow p_1), (p_1 \rightarrow (p_2 \vee p_1)), (p_3 \wedge (p_1 \vee p_2)), (p_2 \leftrightarrow p_1)\}$
 - (b) $U = \{p_1, (p_1 \rightarrow p_2), (p_2 \rightarrow (\neg p_1)), (\neg(p_3 \leftrightarrow p_2))\}$
7. Show that $U \models A$ for a given U and A using semantic tableau.
 - (a) $U = \{(p \rightarrow (q \rightarrow r)), (p \rightarrow q), p\}$ and $A = r$
 - (b) $U = \{(p \rightarrow q), (q \rightarrow r)\}$ and $A = (p \rightarrow r)$
 - (c) $U = \{p, (p \rightarrow q), (q \rightarrow (\neg p)), (\neg(r \leftrightarrow q))\}$ and $A = (r \vee q)$
8. We use $\text{bin}(n)$ to denote the number of binary operators in $U(n)$ and use $\text{neg}(n)$ to denote the number of negation operators in $U(n)$.
 - (a) Let T be a semantic tableau for a propositional logic formula without the bi-implication operator and consisting the other four operators (negation, conjunction, disjunction, implication). For each node n , we define the measure $W_1(n)$ as $2\text{bin}(n) + \text{neg}(n)$. Prove or disprove: If n_1 is a child of n then $W_1(n_1) < W_1(n)$.
 - (b) Let T be a semantic tableau for a propositional logic formula (consisting all the five operators). For each node n , we define the measure $W_2(n)$ as $3\text{bin}(n) + \text{neg}(n)$. Prove or disprove: If n_1 is a child of n then $W_2(n_1) < W_2(n)$.
9. Prove that for every well formed formula A and for every semantic tableau T for A , T is closed iff A is unsatisfiable.